White Paper

Harnessing AI Beyond Chatbots to Innovate Medical Information Offerings

SIMON JOHNS, Director, Medical Information and Marketed Product Safety
LOUISE MOLLOY, Associate Director, Medical Information and Pharmacovigilance



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Introduction

As the pharma industry continues to broaden its global reach to ensure widespread access to critical therapies, each facet of drug development — from manufacturing to commercialization to lifecycle management — must evolve and scale to accommodate growing demand. In the aftermath of the COVID-19 pandemic, needs for customers such as patients and healthcare providers (HCPs) are also changing. Many customers are now looking for more diverse service offerings, including self-service by way of digital tools. As they become more proactive in monitoring their health, customers have grown more likely to reach out to medical information (MI) teams, who oversee the handling of questions requiring the provision of information on medications and their safe, efficacious use. As a result, MI teams are navigating more inquiries than ever.

While inquiries rise, drug companies are working with ever more limited budgets and staff resources. The task of securing the right level of MI expertise in the right location while maintaining budgets is an ever-present balancing act. MI teams have long served as a touchpoint for patients and HCPs to engage with drug companies, and thus, it is critical to meet this growing demand while continuing to provide an efficient, high-quality experience. To strike this delicate balance, many companies are beginning to leverage technological advancements such as artificial intelligence (AI), ChatGPT, and GenAI to offload repetitive tasks, enhance efficiency, and avoid draining vital human resources.

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Navigating a changing MI landscape

As pharmaceutical companies enable improved global access to their therapies, call centers are being inundated with a growing number of inquiries, and volume is not their only challenge. Globalization has led to inquiries across languages, regulatory jurisdictions, and regions. Furthermore, as oncology products, biologics and cell and gene therapies continue to surge across the industry, MI agents must be able to navigate complex inquiries by staying well-versed in advanced medical processes. These factors require MI agents to have a diverse and adaptable skillset to ensure consistent and accurate responses across a global customer network.

As a result of these compounding factors, the cost of acquiring and supporting suitably skilled MI agents has grown. In addition to being placed in the most appropriate jurisdiction, agents need a manageable bandwidth that allows them to engage with patients and HCPs thoughtfully as they are often the primary inbound communication channel for drug companies. Because of this vital communication gateway, drug companies must also consider how patient and HCP needs have evolved.

In a self-service-driven marketplace, patients and HCPs are often looking to engage with MI via digital tools such as online portals, automated chat services, and mobile apps. However, to successfully leverage digital technology to improve the customer experience, companies must ensure the accuracy and consistency of information provided digitally. Embedded alongside the growing need for more digitally accessible tools is a level of cultural uncertainty around AI usage that must be mitigated via thoughtful legislation and rigorous procedures.

In April 2021, the EU passed the EU AI Act, which provides a comprehensive legal framework for implementing AI systems. This legislation places the onus on companies to be proactive in understanding how the framework applies to them to ensure any AI systems they develop are compliant, ethical, and safe.

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One provision of the EU AI Act categorizes AI systems at different risk levels based on their function. Given that MI AI systems will have access to highly sensitive HCP and patient data, it is likely they will be classified as high risk. As a result, these systems must be heavily vetted, tested, documented, and monitored continuously to ensure the output is safe, unbiased, and ethical prior to activation, with any responses to questions based only on drug company approved content. Perhaps most importantly, these systems need to establish human and AI synergy to ensure that the future of MI is rooted in a blend of AI efficiency and human expertise and judgement.

Leveraging strategic developments and technological advances

Though conducting a major process overhaul is never easy, the traditional human-only approach to MI is no longer sufficient to meet current demands. As drug companies recognize this, their teams are seeking to employ innovative, robust solutions that incorporate AI to deliver reliable information to inquirers with high efficiency, while still being focused on excellent customer engagement. This requires retooling the human-only model to incorporate an augmented AI-support structure, which complements the nuanced work done by MI specialists.

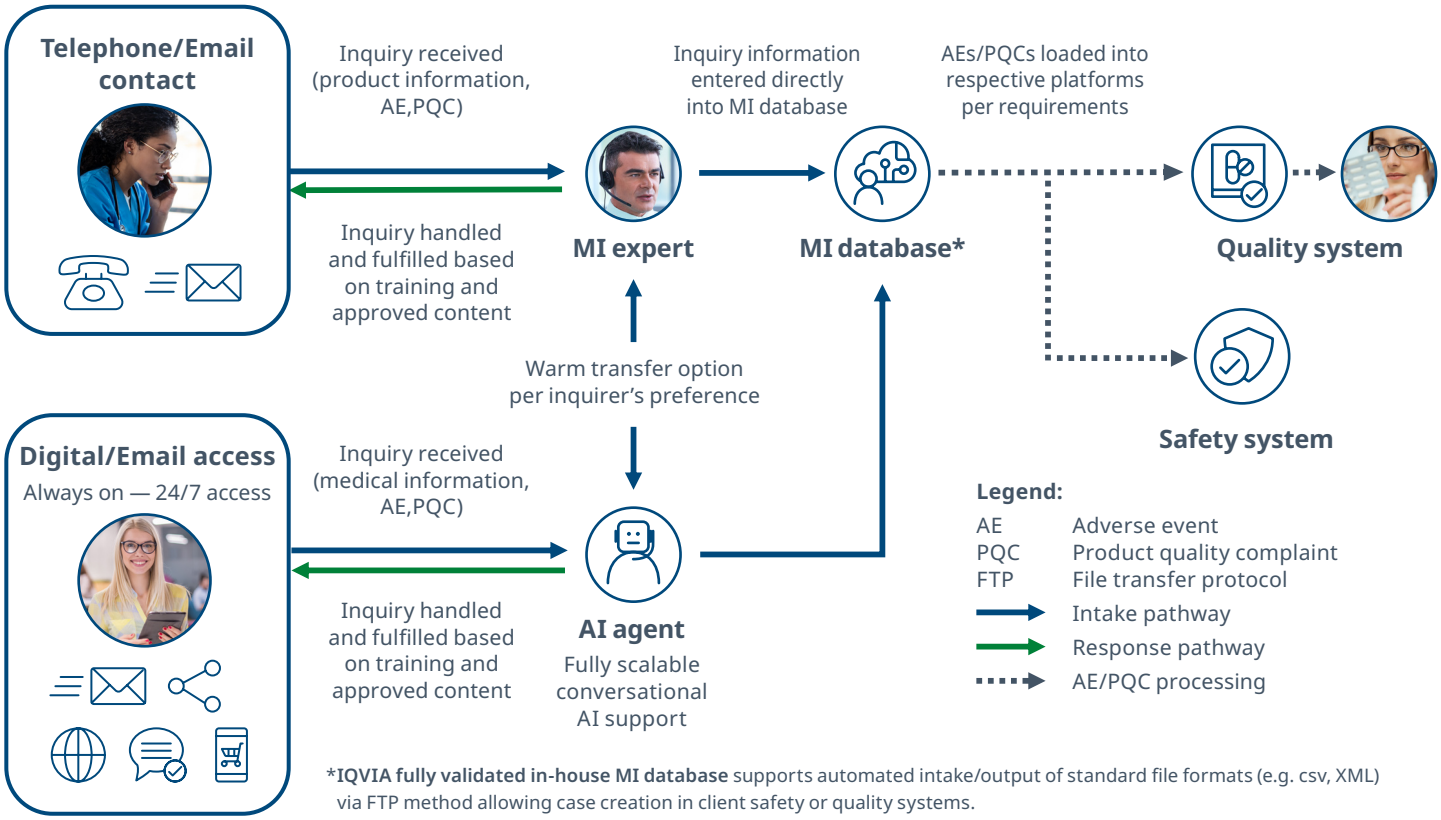
The benefits of implementing AI into MI workflows are twofold. First and foremost, it minimizes delays by providing inquirers with rapid access to reliable information and, hopefully, as a result, mitigating their

need for follow-up calls. Secondly, it offloads tedious administrative tasks to AI, freeing up time for expert staff to focus on providing high-quality service; this helps to alleviate burnout and ensure retention. The realization of AI solutions also allows for an enhanced suite of services to accommodate a wider range of customer preferences and availabilities. As medicine itself moves towards more personalized solutions, it is beneficial for MI to follow suit by offering more personalized provisions that align with customer needs, whether they want a rapid digital solution or human touchpoints.

Figure 1 demonstrates a practical combined human/AI workflow for guiding a customer inquiry through to fulfillment, as well as enabling 24/7 support. A key component of this workflow is that inquirers always have the option of being transferred from an AI agent to a human agent, depending on their preference. As chatbot mechanisms continue to evolve beyond simple yes/no or dropdown list approaches, they are becoming more sophisticated conversational AI models that effectively engage with human inquirers. Thus, AI and human agents can work in a complementary fashion to satisfy customer inquiries. Furthermore, this model is fully scalable for unpredictable volume spikes; the AI model is on-hand to assist human agents if inquiry volumes become unmanageable.

In a high-volume spike, AI will feed received data into the MI database in real time, combined with continuous provision of up-to-date, accurate material to human agents as they handle inquiries. Other types of automation include the automatic transfer of adverse events (AEs) and product complaints (PQCs) to their requisite systems, which helps maintain compliance amid spikes. Ultimately, the integration of AI and data analytics into MI enhances capabilities while tailoring a rapid, more personalized experience for HCPs and patients.

Figure 1: A combined human/AI contact center model



***IQVIA fully validated in-house MI database** supports automated intake/output of standard file formats (e.g. csv, XML) via FTP method allowing case creation in client safety or quality systems.

Incorporating data analytics and real-world examples

Drug companies looking to implement AI solutions in their MI initiatives would be well-advised to outline a technology road map towards their goals. To ease the transition to a combined human/AI model, consider how others have leveraged AI to guide future initiatives. There is already a real-world precedent for applying AI across the following MI initiatives:

- Supporting higher inquiry volumes while minimizing resource impact.
- Creating engagement options to reduce inquiry abandonment and improve customer satisfaction.
- Providing inquirers with the choice between an AI agent or a human specialist.

When considering where AI could complement your MI workflows, assess which simple, repetitive tasks might be automated while still maintaining a level of human oversight. To ensure that AI delivers safe, effective, accurate, and approved information to patients and HCPs, it is vital to put guardrails in place to prevent AI from "hallucinating" or gathering a response from non-approved internet sources. By continuing to have human oversight and governance, you avoid such problems.

As information continues to be amassed via MI inquiries, this data can be harnessed using data analytics and AI capabilities to create real-time MI dashboards that assess significant amounts of data to convey emerging inquiry trends across therapeutic areas and markets. From there, data analytics will inform your AI strategy, i.e., if there is an influx of simple inquiries within a particular market, AI capabilities can be applied there while human agents are assigned to more complex activities. Data analytics can also identify opportunities for further process optimization. One example of this is using data to ensure that existing content is suitable for inquiry fulfillment or HCP/patient education, i.e., determining whether approved content is fit for purpose or could be further optimized.



From an AI optimization perspective, AI is capable of automating call transcriptions to text in the local language, as well as providing an English version if required. It can also conduct real-time sentiment analysis to determine how the inquirer is behaving over the phone, and how the agent is handling the inquiry. Consider, are they frustrated with the response time? Are they repeating the same question multiple times? By gathering this information, a team can better assess how the agent is performing and how calls could be improved going forward. AI can also provide live agent assistance in which approved content is provided in real time to a human agent to ensure they have pertinent information on hand. This prevents them from having to place an inquirer on hold to find the necessary resources and is yet another example of how leveraging AI can optimize the productivity, timeliness, and accuracy of an MI workflow.

Crafting the ideal MI approach

Across the MI space, more and more drug companies are turning to AI to accommodate a growing number of inquiries. Understanding your existing MI inquiry volume and recognizing how your team is equipped to handle ever-changing requirements, demands, and patient and HCP expectations is a critical step to ensuring customer satisfaction and employee retention in the current MI climate. The better you have defined your MI needs, the better positioned you are to design a combined human/AI service model geared towards patient and HCP support.

IQVIA

IQVIA, a leading global MI solutions provider, began working with customers to integrate AI as part of MI as early as 2020, when huge fluctuations in inquiry volumes started to take hold. To accommodate customers in a cost-effective fashion, IQVIA developed the combined human and conversational AI agent model to handle MI inquiries. After being rolled out in 2021, this model was successfully executed across several large-scale customer projects and, as a result, IQVIA was recognized by the American Business Awards with a Gold Stevie Award for the implementation of an augmented AI solution. At present, this work is still recognized as one of the few successful rollouts of AI in MI systems.

When ChatGPT emerged in 2022, IQVIA was already several years ahead of the AI curve. In our ongoing commitment to creating customer-focused solutions, IQVIA works with each client to build a customized AI roadmap to identify how AI might offer process optimization to improve the customer journey and maintain cost-effective offerings.

About the authors



SIMON JOHNS
Director, Medical Information and Marketed Product Safety, IQVIA

Simon Johns manages global MI projects focused on process optimization and technology enablement to drive enhanced efficiency and customer engagement. Johns is a member of the European DIA Medical Information and Communications Training Team, advising pharmaceutical companies on best industry practices, innovation and automation. He speaks regularly on topics ranging from combined human and AI conversational agent models for MI to the benefits and increased value of integrating MI and pharmacovigilance. Located in England, Johns has over 25 years of experience supporting global pharmaceutical customer projects.



LOUISE MOLLOY
Associate Director, Medical Information and Pharmacovigilance, IQVIA

Louise Molloy manages operations for the Medical Information and Pharmacovigilance team at IQVIA, focusing on optimizing processes and leveraging technology to enhance efficiency and compliance. Molloy is an integral part of the global team, advising pharmaceutical companies on best practices, innovation, and automation in MI. She frequently speaks on topics such as integrating AI in Medical Information systems and the evolving landscape of pharmacovigilance. Located in Spain, Molloy brings over 15 years of experience in the pharmaceutical industry.

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